Midterm Presentation

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Monday 15th September, 2025

1. Goal
2. FEM Introduction
3. Project and Timeline
4. Implementation
5. Next Steps

Outline

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Goal

Full Maxwell solver in IPPL

- FDTD solver exists, only 2nd order
- Finite Elemente Methods Maxwell solver
  - Allows for higher accuracy
  - Matrix free implementation allows for similar memory usage
  - Only increases the computational load on the GPU

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Finite Element Method (FEM)

Consider the strong form of a PDE (example: Poisson equation):

$$-\Delta u = f \quad \text{in } \Omega, \quad u = 0 \text{ on } \partial\Omega$$

The Variational (weak) form: Find $u \in V = H_1^0(\Omega)$ such that

$$a(u, v) = \int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx = l(v) \quad \forall v \in V$$

FEM approximation:

- Discretize the domain in to a mesh $\mathcal{M}$ with elements K: $\overline{\Omega} = \bigcup_{e=1}^{N_E} K_e$
- Choose finite-dimensional subspace $V_h \subset V$
- Find $u_h \in V_h$ such that: $a(u_h, v_h) = l(v_h) \quad \forall v_h \in V_h$

This leads to the system of equations $A\mathbf{x} = \mathbf{b}$ that can be solved matrix free

Lagrange Finite Elements

- V_h can be chosen as the **Lagrange space** of degree p:

$$\mathcal{S}_p^0(\mathcal{M}) = \{v_h \in C^0(\overline{\Omega}) : v_h|_K \in \mathbb{P}_p(K_e),\ \forall K_e \in \mathcal{M}\}$$

with the polynomials of degree $\leq$ p on element K:

$$\mathbb{P}_p(K) := \{x \in K \mapsto \sum_{\alpha \in \mathbb{N}_0^d, |\alpha| \leq p} c_\alpha x^\alpha,\ c_\alpha \in \mathbb{R}\}$$

- The functions in V_h are fully determined by their values at a finite set of degrees of freedom (DOFs), associated with vertices, edges, faces and volumes.
- A basis for V_h is given by $\{b_N^i\}_{i=1}^N$, the Lagrange basis functions satisfying $b_N^i(\mathbf{x}_j) = \delta_{ij}$, $N =$ number of DOFs

From Variational Problem to Linear System

We discretize u:

$$u_h = \sum_{j=1}^{N} \mu_j b_N^j$$

Inserting this into the variational formulation:

$$a(u_h, v_h) = l(v_h) \quad \forall v_h \in V_h$$

$$\Rightarrow \quad \sum_{j=1}^{N} \mu_j a(b_N^j, b_N^i) = l(b_N^i), \quad i = 1, \ldots, N$$

which can be written as the linear system

$$A\mu = \mathbf{b}$$

where$A_{ij} = a(b_N^j, b_N^i), \quad b_i = l(b_N^i), \quad \mu = (\mu_1, \ldots, \mu_N)^T$

Matrix-Free FEM Solver (in IPPL)

In the matrix-free FEM approach, we avoid assembling the stiffness matrix A explicitly.

- Righthand side information provided on ippl::Field
- Variational formulation defining the operator A

Workflow:

1. **FEM Interface:**
   - Projects the RHS onto the FEM space, now called b.
   - Provides a function $evaluateAx()$ to evaluate $A \cdot x$ matrix-free.
2. **CG:** Solves $A\mu = b$ for the coefficient vector μ using $evaluateAx()$.
3. **FEM Solution:** The solution $u_h \in V_h$ is obtained via:

$$u_h(x) = \sum_{j=1}^{N} \mu_j b_h^j(x)$$

4. **Transfer Back:** The solution is transfered back from the FEM space to the ippl::Field

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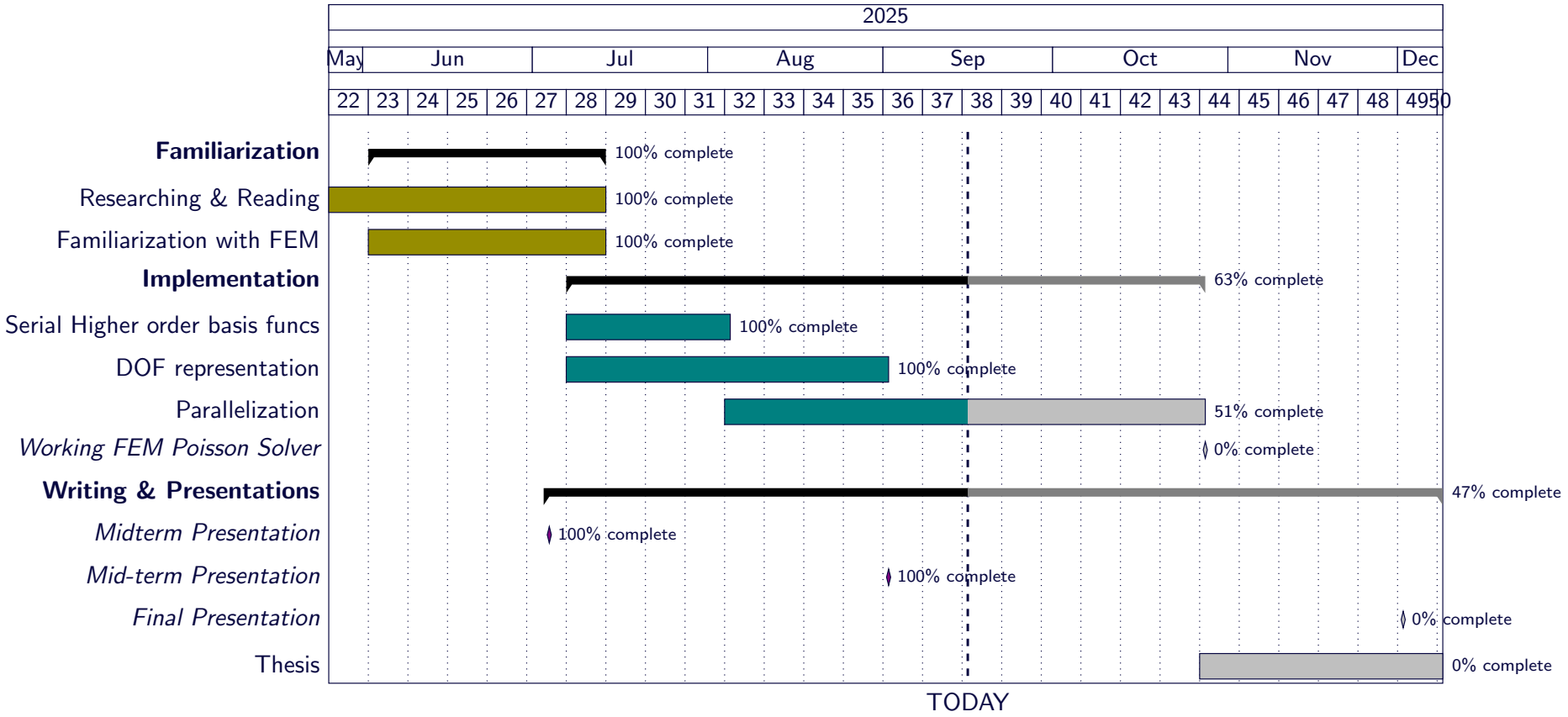
What is already implemented

- A basic **FEM framework** has been introduced:
  - Matrix-free, using structured hexahedral grids
  - Supports first-order Lagrange finite elements
  - `FEMPoissonSolver` provides a working example
- A separate project has implemented:
  - First-order Nédélec finite elements for Maxwell equations
  - The `FEMVector` used to represent DOF
  - `FEMMaxwellDiffusionSolver` provides a working example

My Task

- Introduce higher order basis functions on top of the Lagrange space to enable p-refinement.
- Software design of the general higher-order elements in IPPL.
- Correctness studies, performance analysis and scaling studies of the implemented code.

Timeline


2025
May
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Dec
22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50
Familiarization
100% complete
Researching & Reading
100% complete
Familiarization with FEM
100% complete
Implementation
63% complete
Serial Higher order basis funcs
100% complete
DOF representation
100% complete
Parallelization
51% complete
Working FEM Poisson Solver
0% complete
Writing & Presentations
47% complete
Midterm Presentation
100% complete
Mid-term Presentation
100% complete
Final Presentation
0% complete
Thesis
0% complete
TODAY

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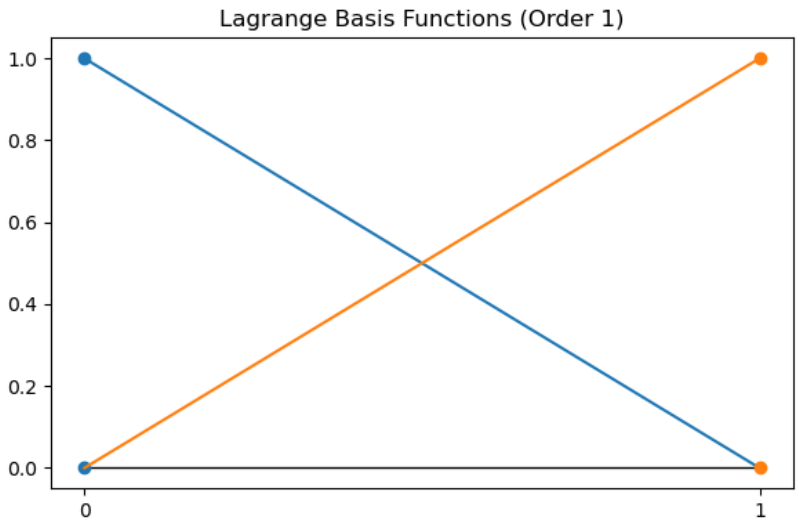
Lagrange Basis Functions in 1D

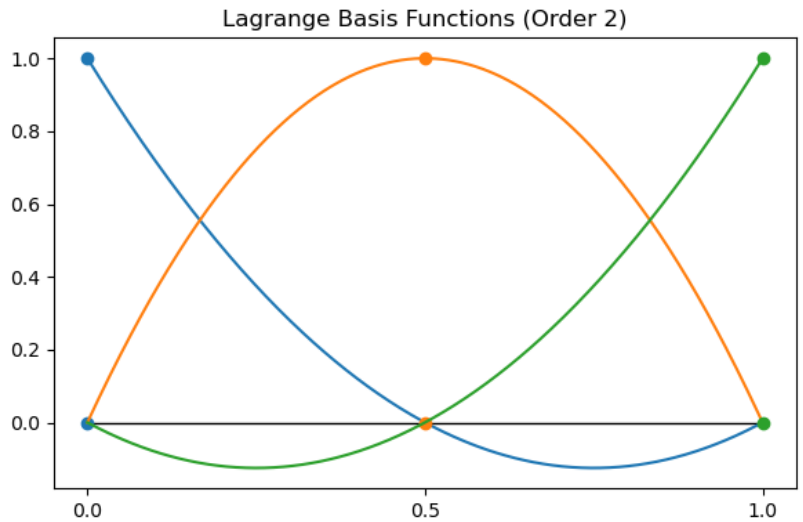
Lagrange basis functions are defined such that $b_N^i(x_j) = \delta_{ij}$ for $i, j = 1, \ldots, N$. The Lagrange basis function $b_N^i(x)$ on interpolation points (DOFs) $x_1, \ldots, x_N$ is given by:

$$b_N^i(x) := \prod_{\substack{j=1 \\ j \neq i}}^{N} \frac{x - x_j}{x_i - x_j}, \quad i = 1, \ldots, N$$

Their gradient is:

$$\frac{d}{dx} b_N^i(x) = \sum_{\substack{k=1 \\ k \neq i}}^{N} \left(\prod_{\substack{j=1 \\ j \neq i, j \neq k}}^{N} \frac{x - x_j}{x_i - x_j} \cdot \frac{1}{x_i - x_k} \right)$$


Lagrange Basis Functions (Order 1)
1.0
0.8
0.6
0.4
0.2
0.0
0
1


Lagrange Basis Functions (Order 2)
1.0
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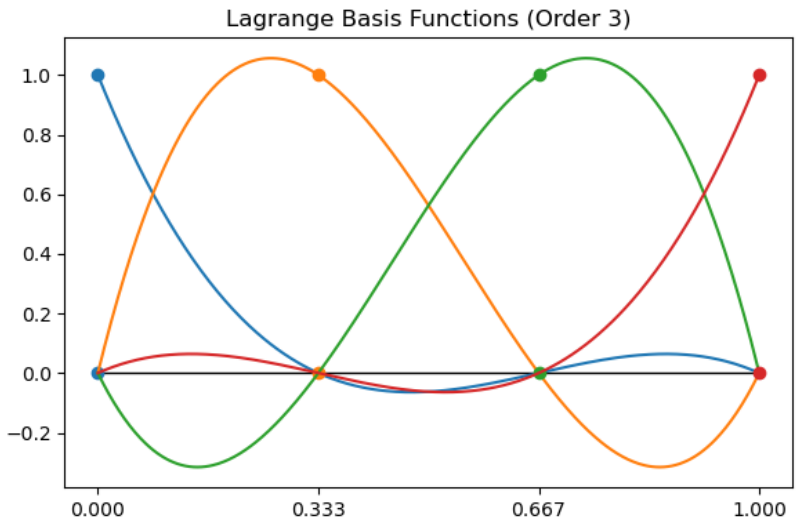
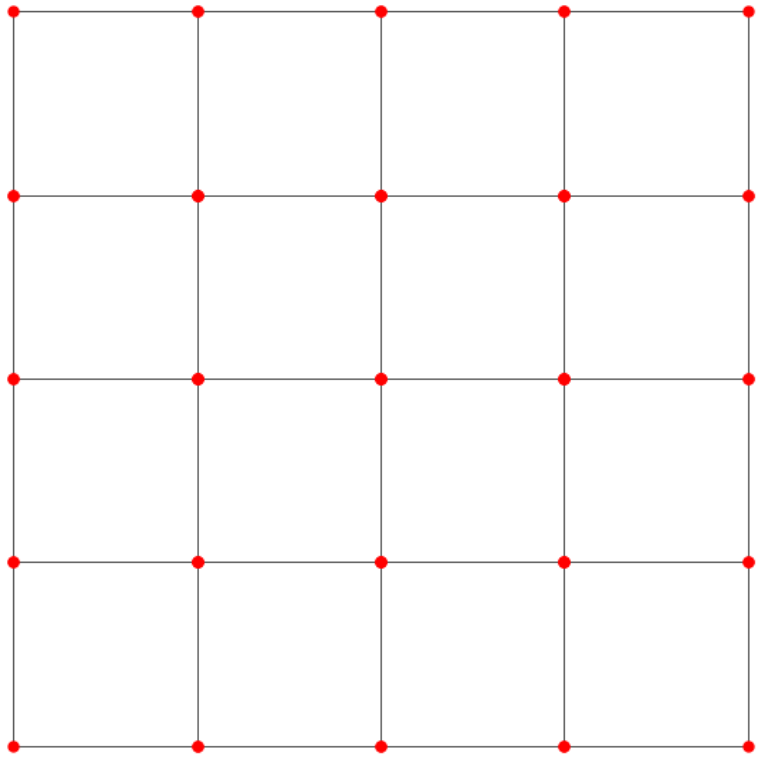
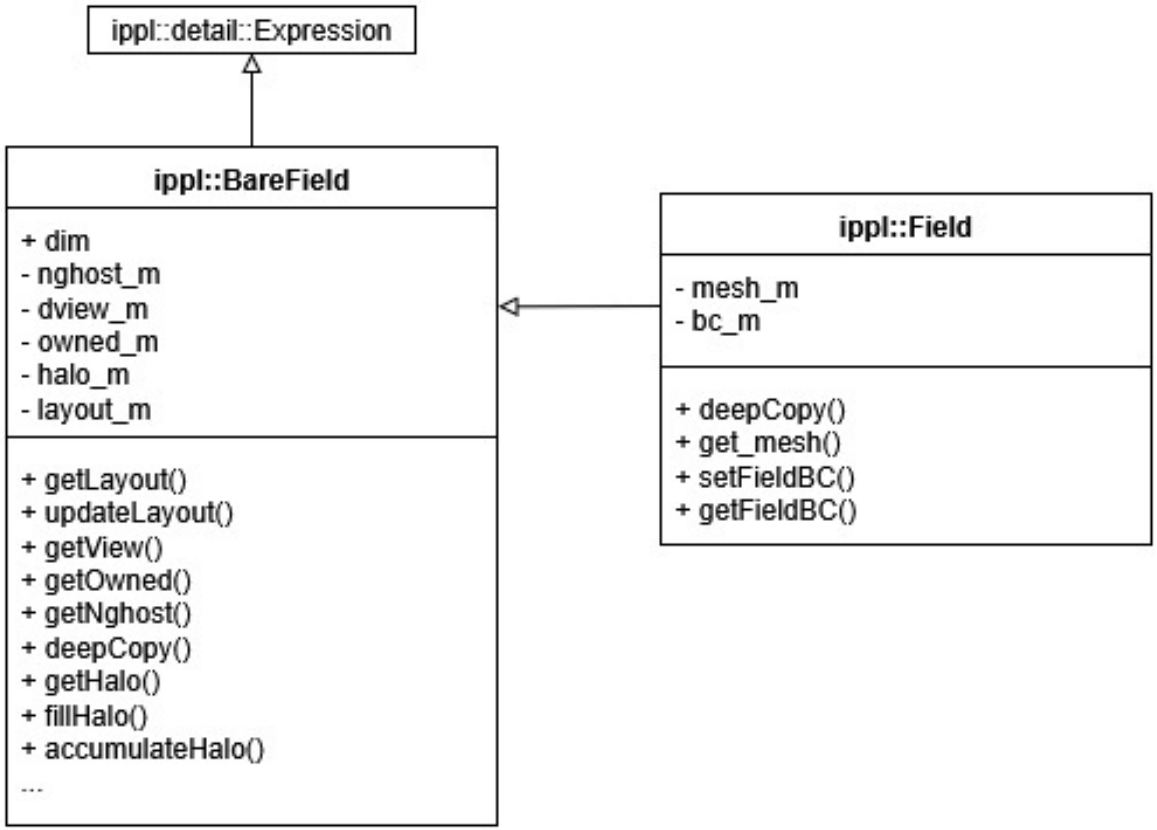
0.6
0.4
0.2
0.0
0.0
0.5
1.0

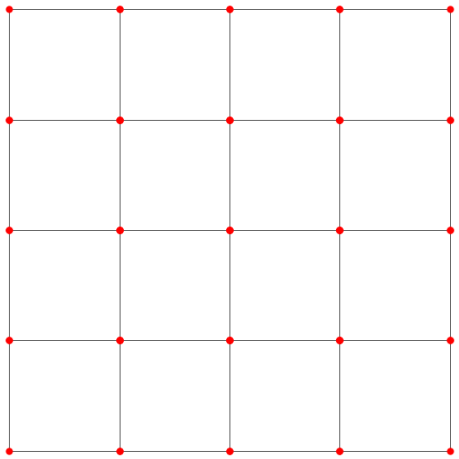

Lagrange Basis Functions (Order 3)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.000
0.333
0.667
1.000

DOF Representation first order Lagrange

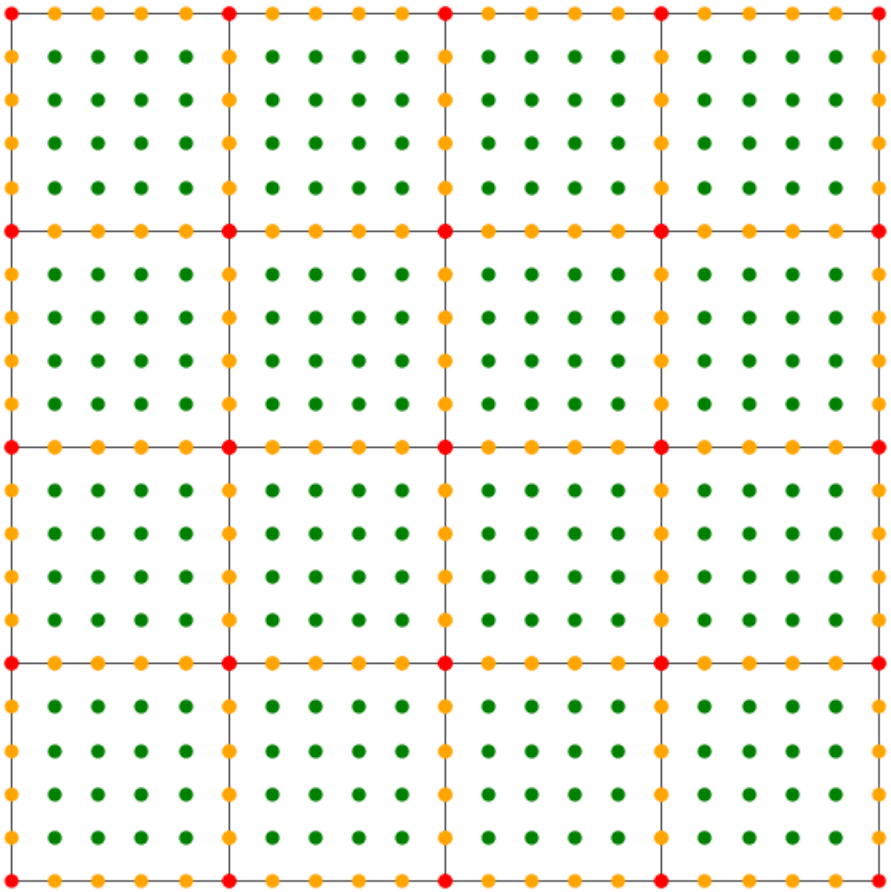


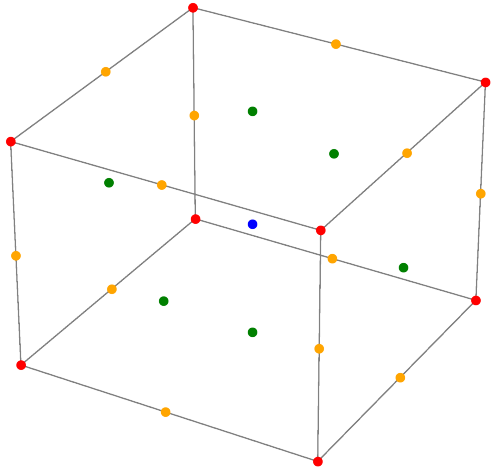
DOF Representation first order Lagrange



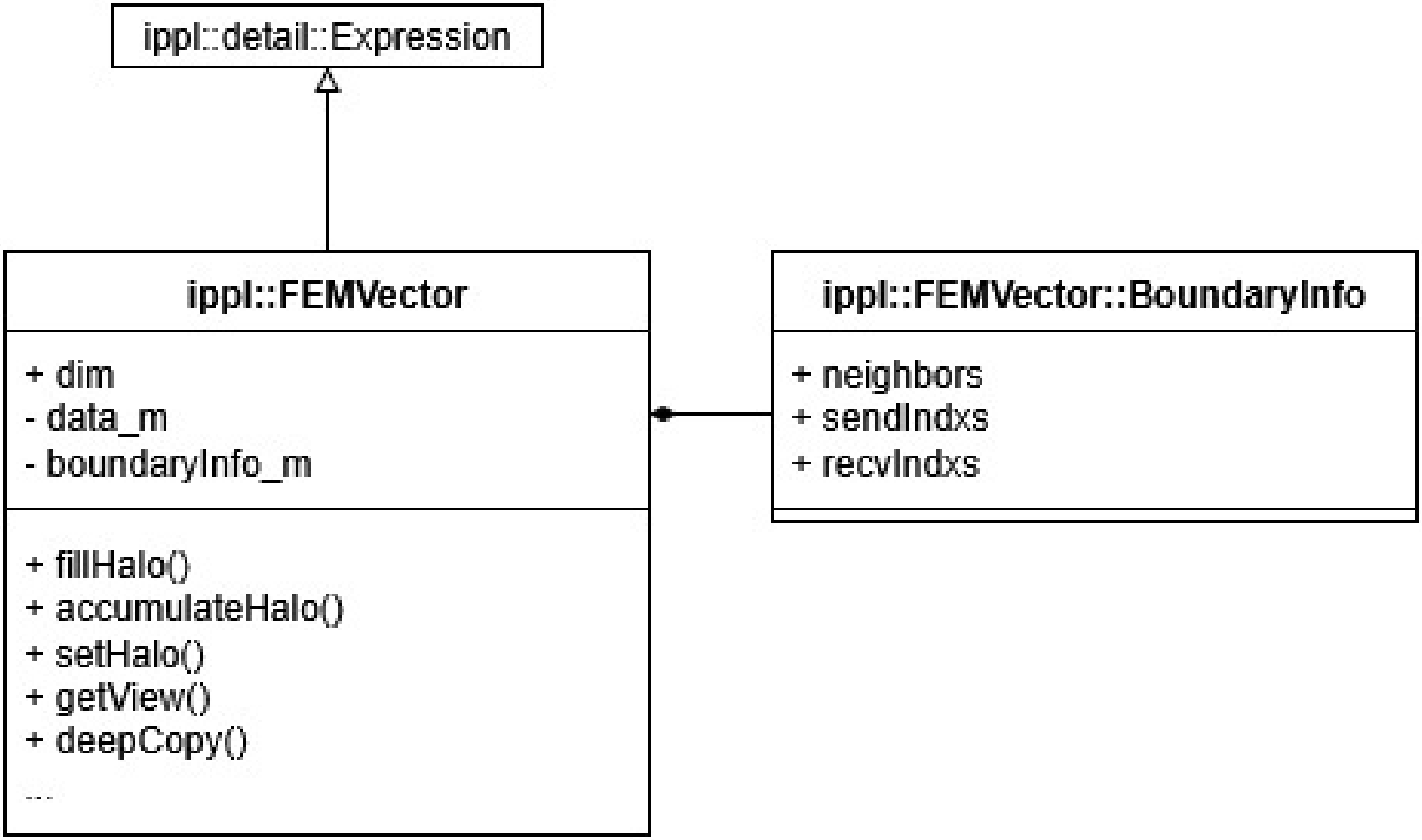


Representing additional DOFs





The FEMVector

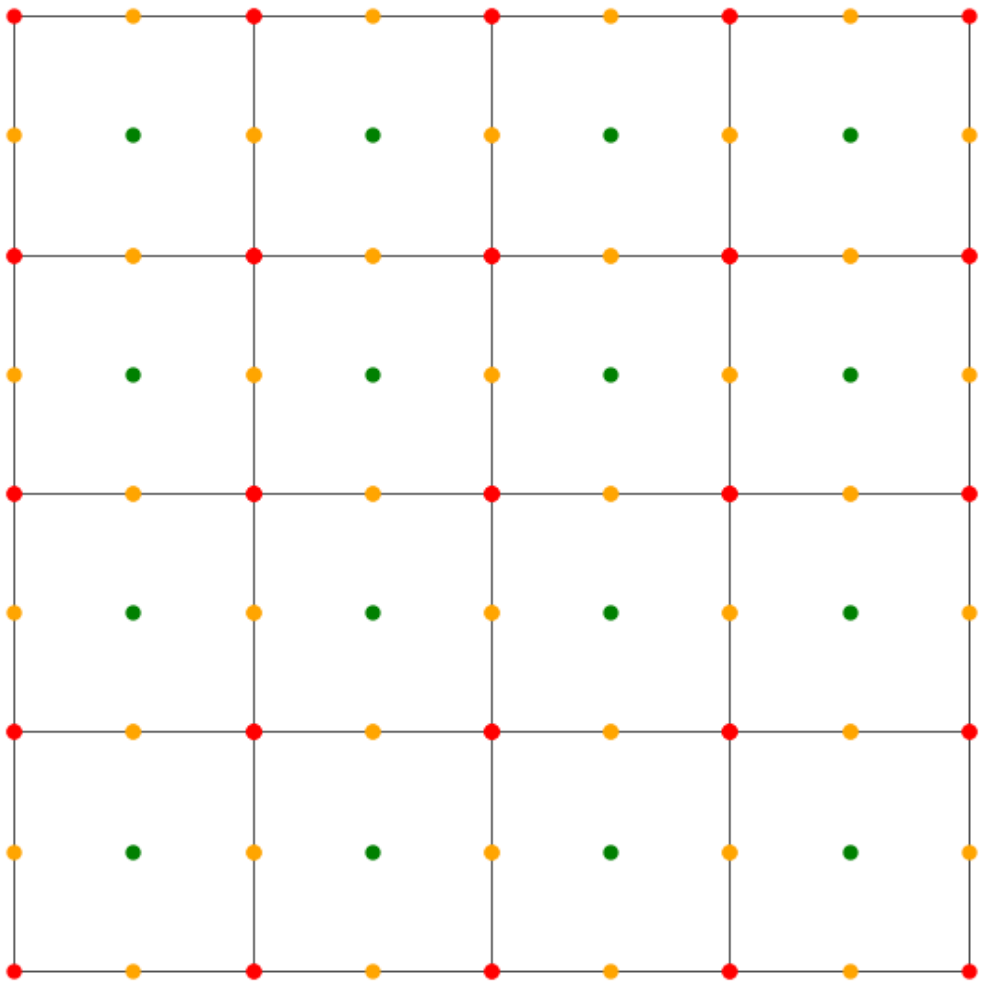


The `FEMVector`

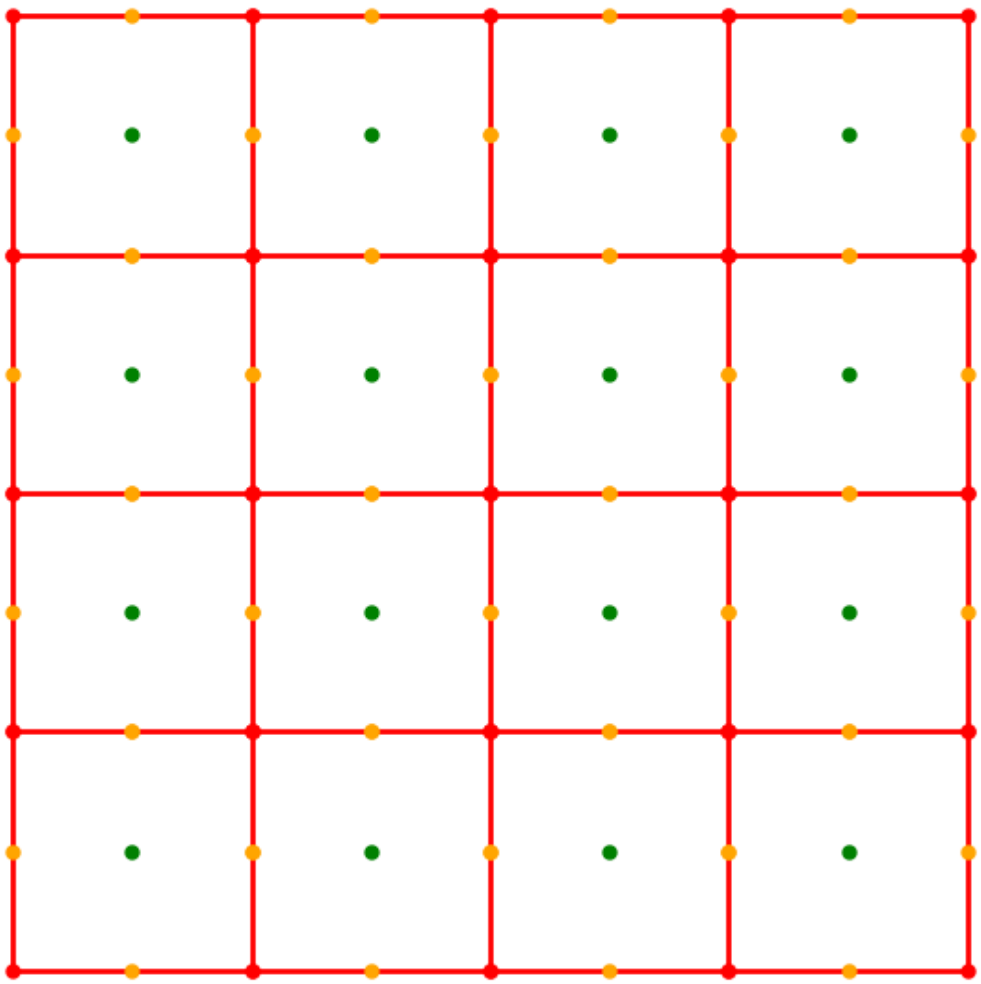
- Represents all DOFs using a 1D `Kokkos::View`.
- Enables parallelization via MPI halo exchange:
  - Based on index lists: `neighbors`, `sendIdxs`, `recvIdxs`.
  - Uses `fillHalo()`, `accumulateHalo()`, and `setHalo()`.
- **Advantages**:
  - Works for all FEM spaces.
  - Independent of mesh geometry and layout.
- **Drawback**:
  - Requires custom DOF mapping and halo setup for each FEM space.

Is there an alternative data structure which simplifies the halo exchange?

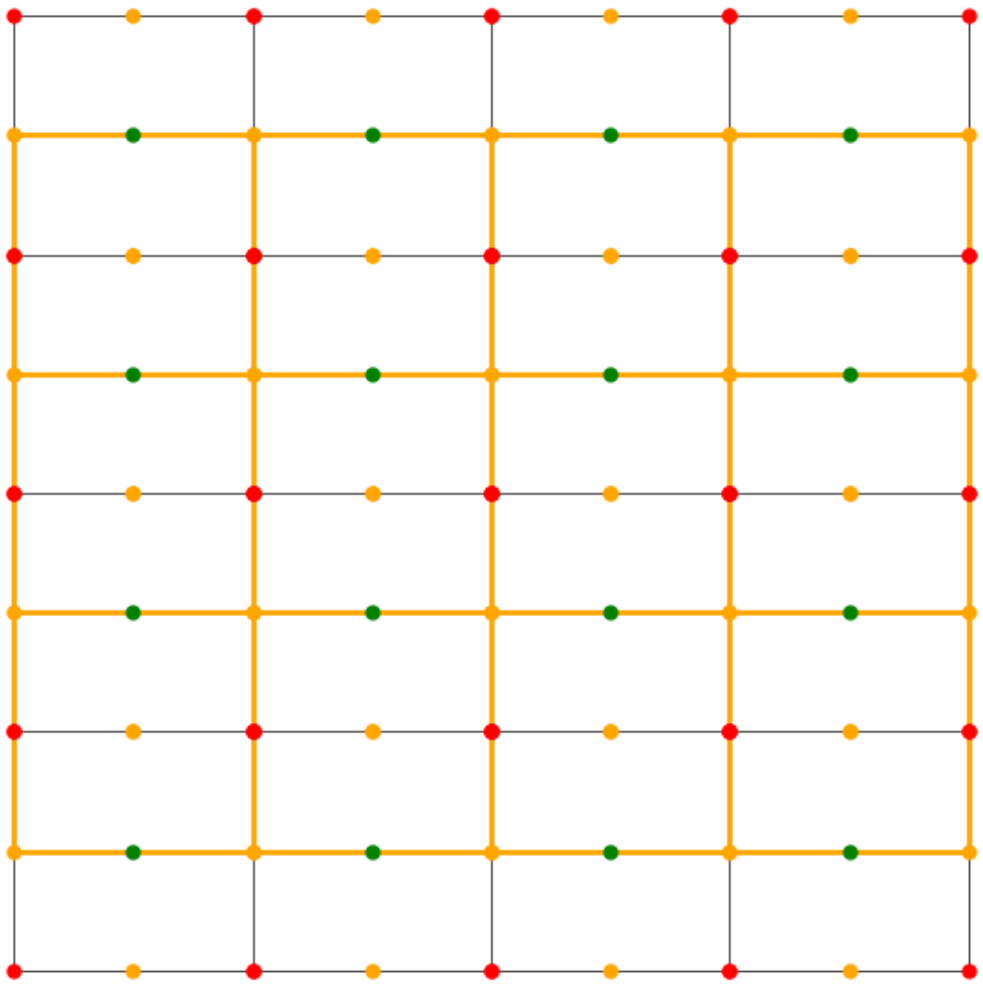
Alternative representation of additional DOFs



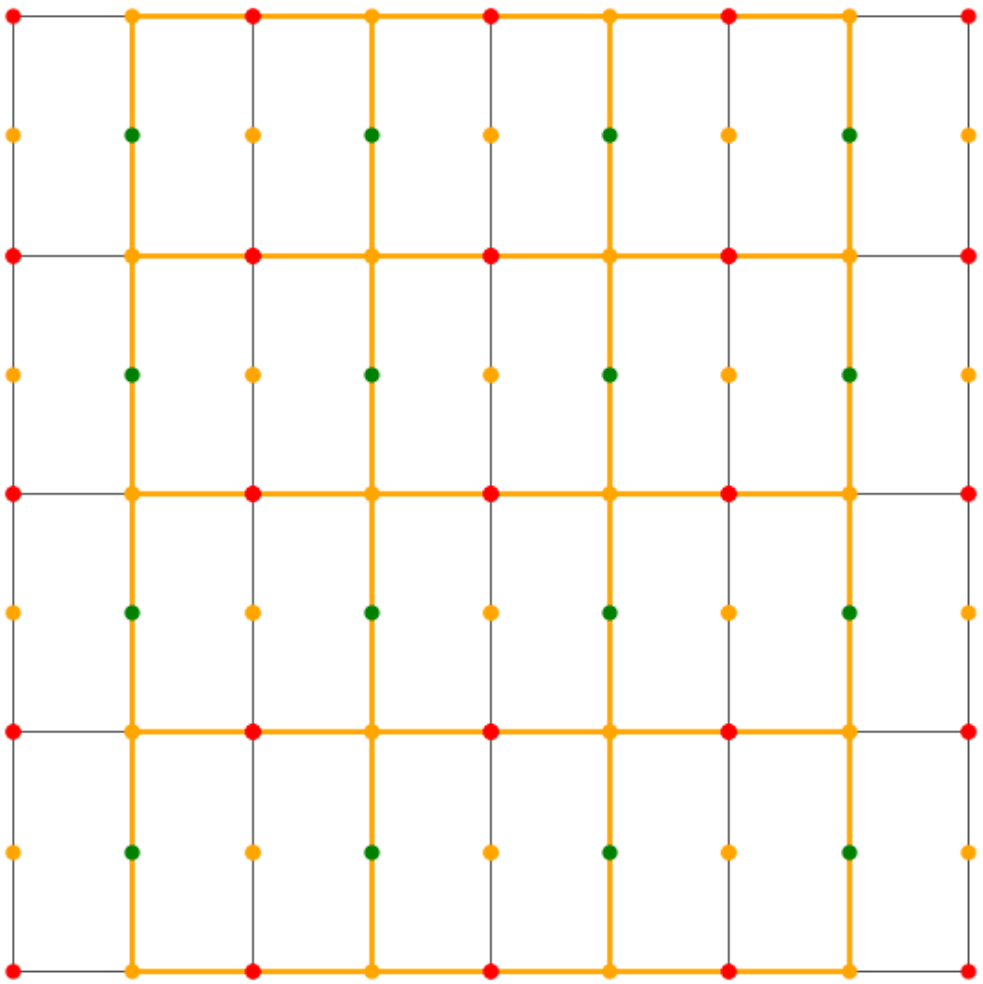
Alternative representation of additional DOFs



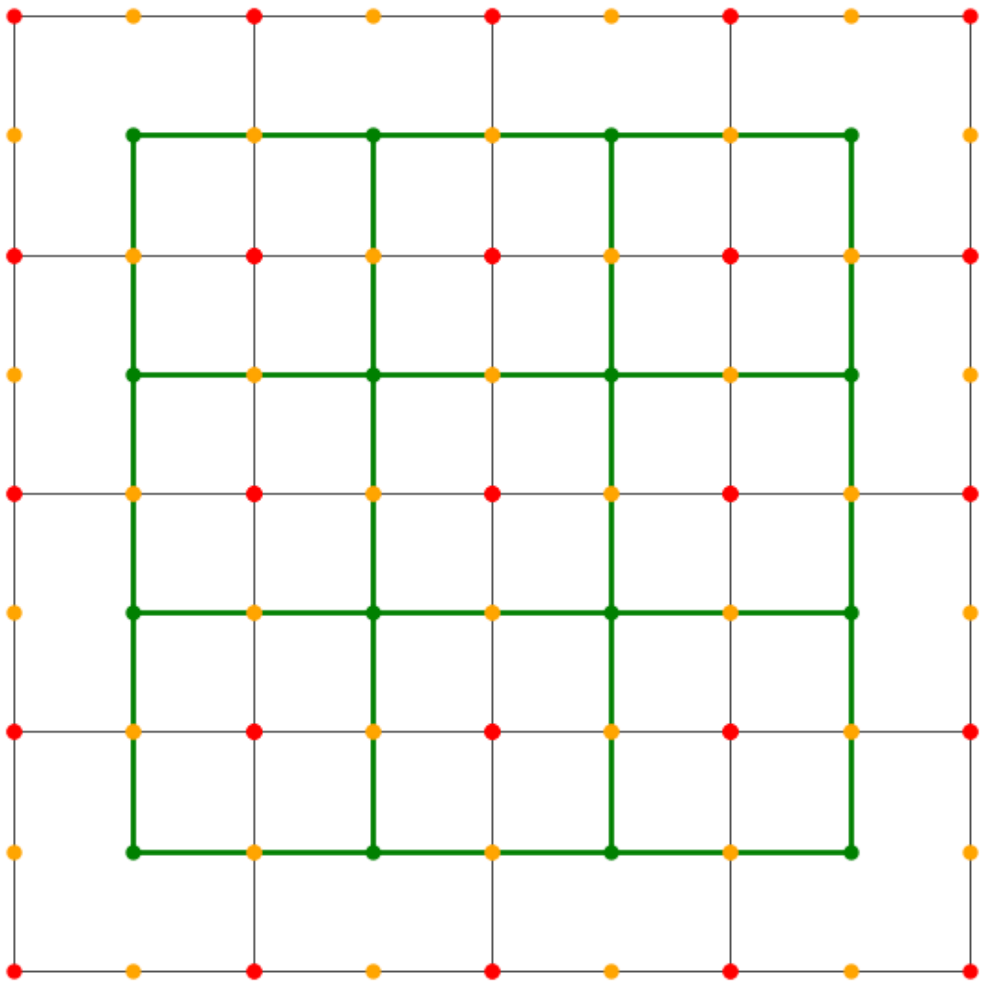
Alternative representation of additional DOFs



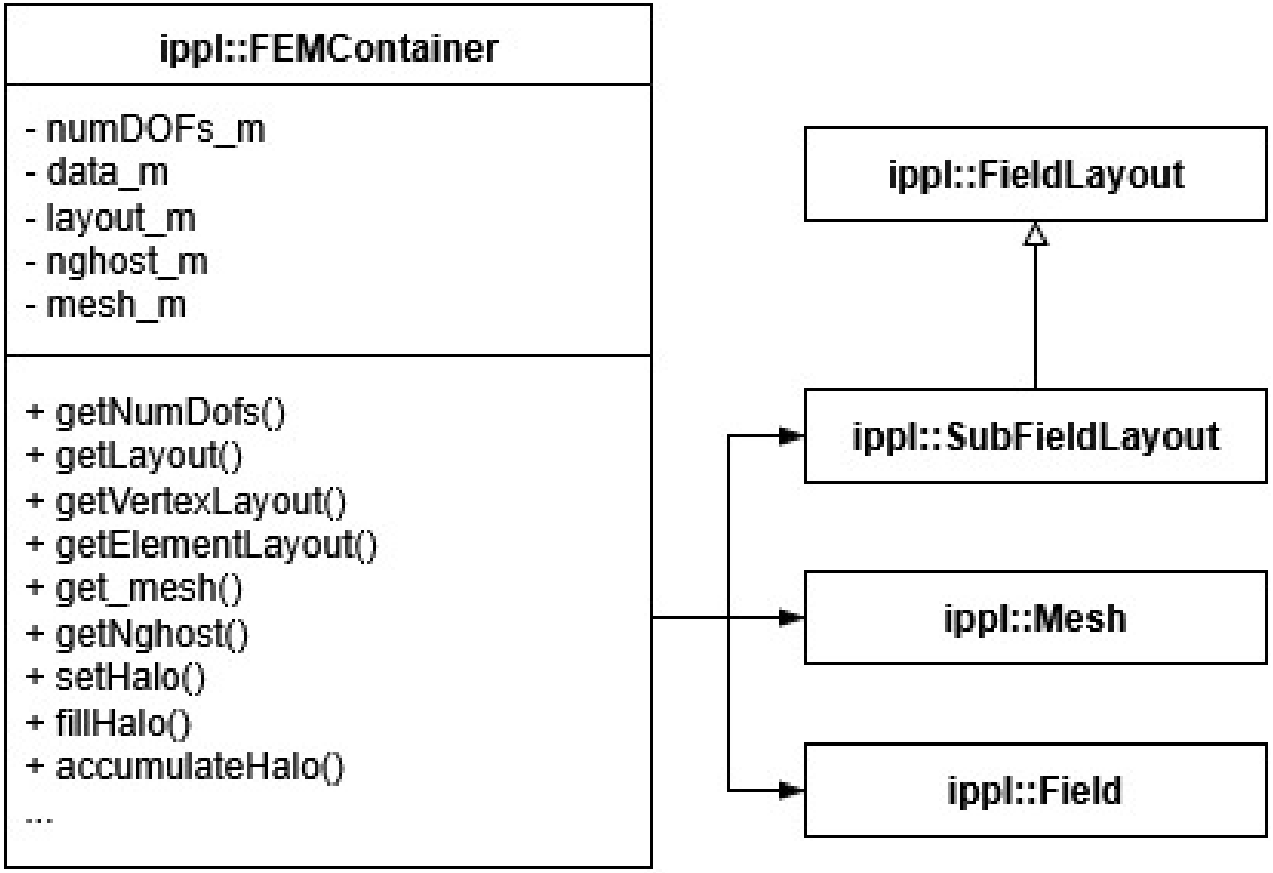
Alternative representation of additional DOFs



Alternative representation of additional DOFs



The FEMContainer



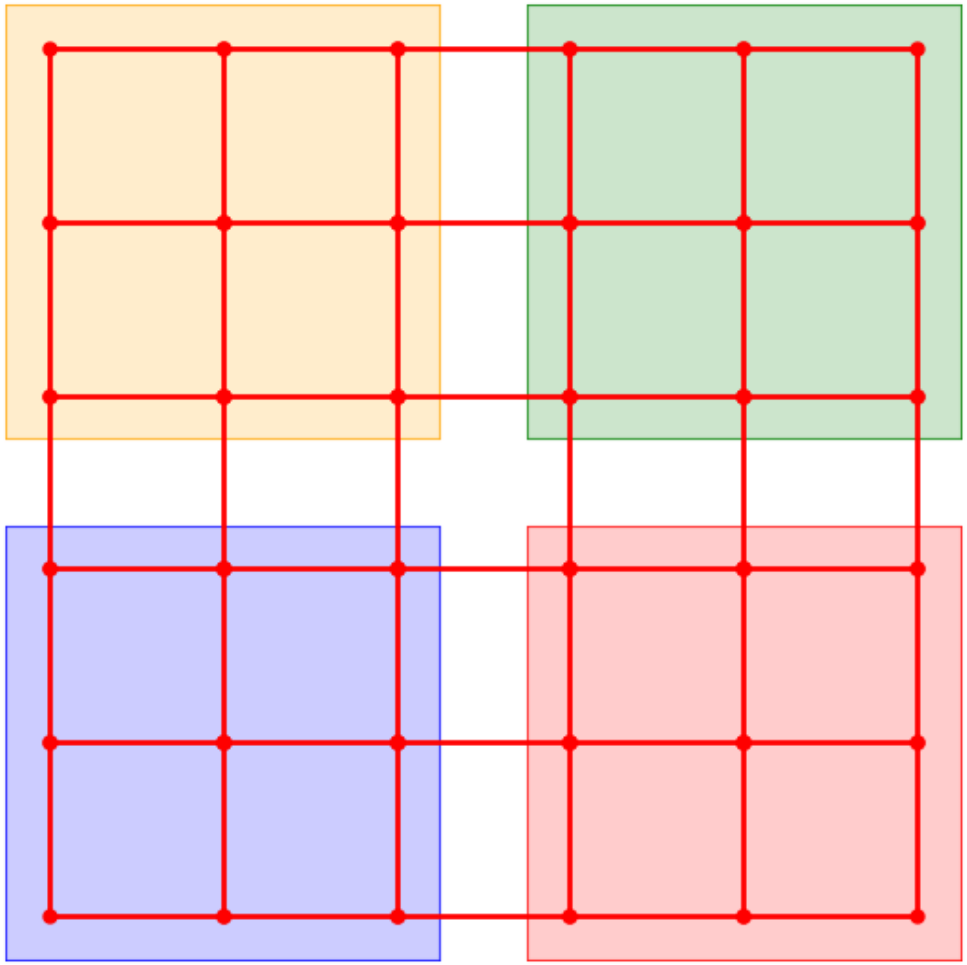
The FEMContainer

- Represents DOFs on multiple ippl::Fields for each element type and direction
- Uses via MPI parallelized halo exchange from ippl::Field
- **Advantages**:
  - Works for all FEM spaces.
  - Halo exchange works for all FEM spaces and all orders

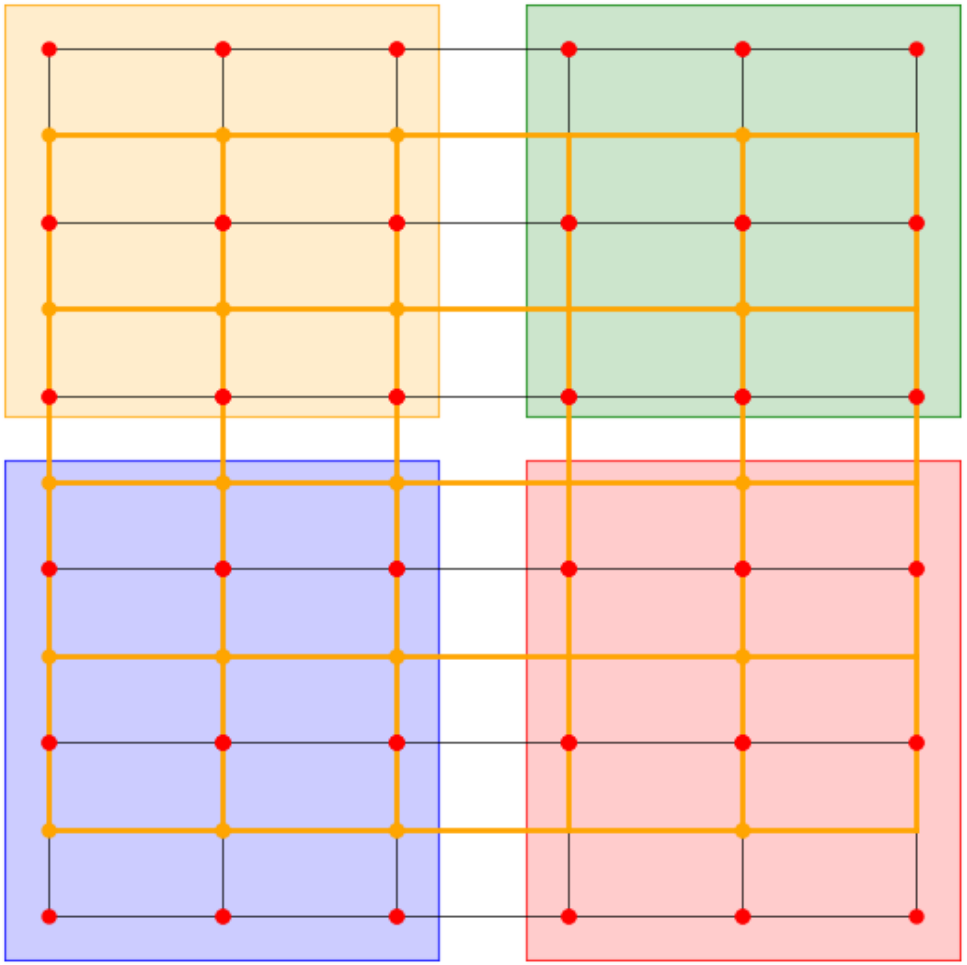
The SubFieldLayout

How to coordinate parallelization of the differently sized fields in the FEMContainer?

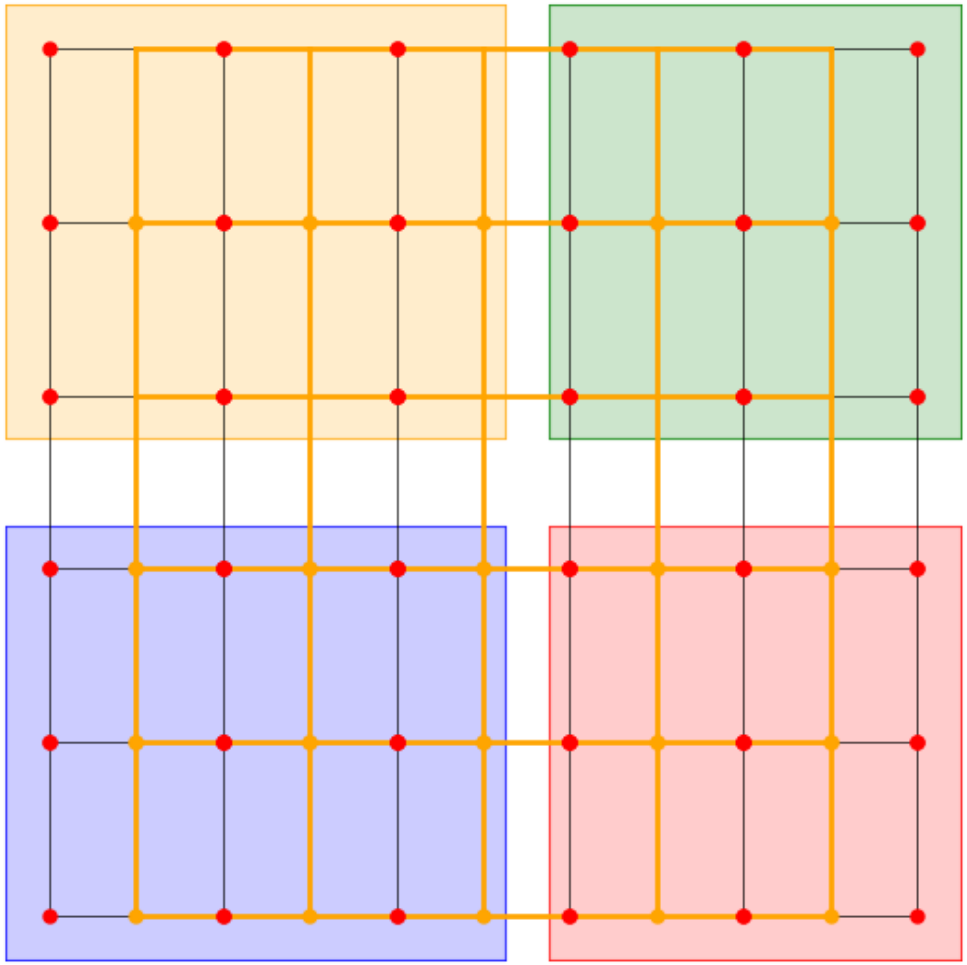
The SubFieldLayout



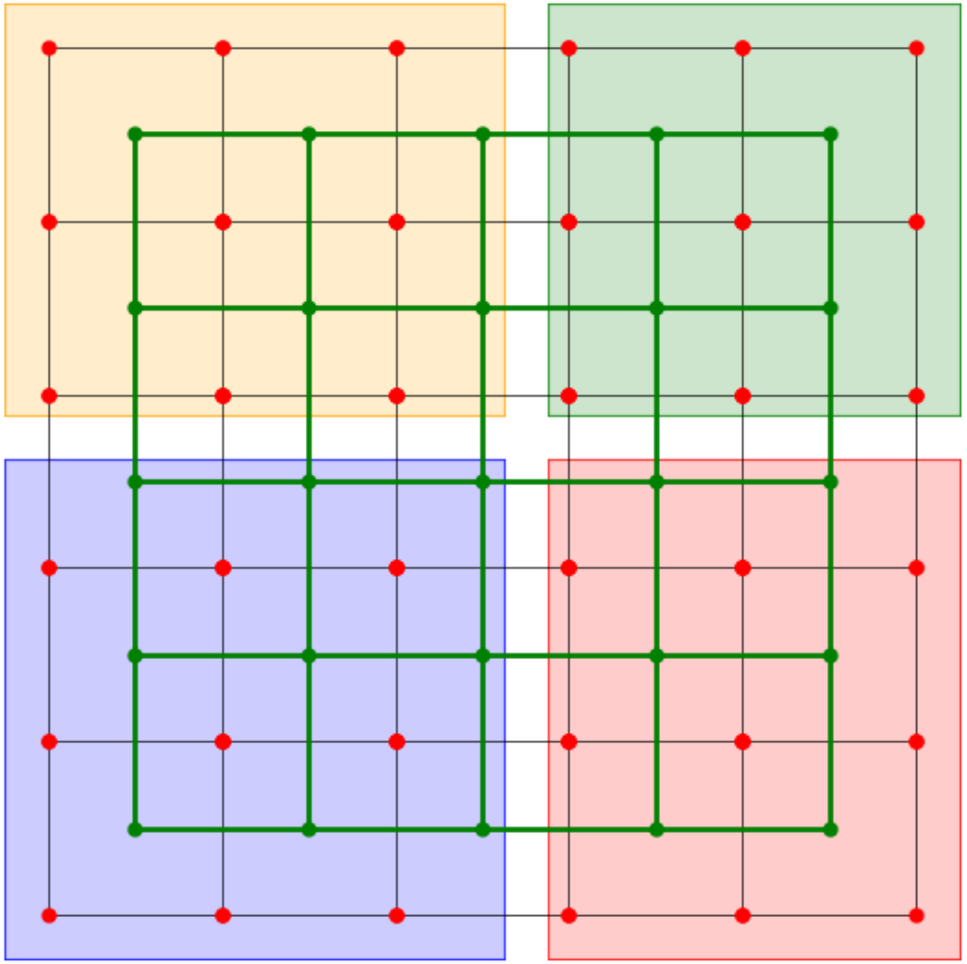
The SubFieldLayout



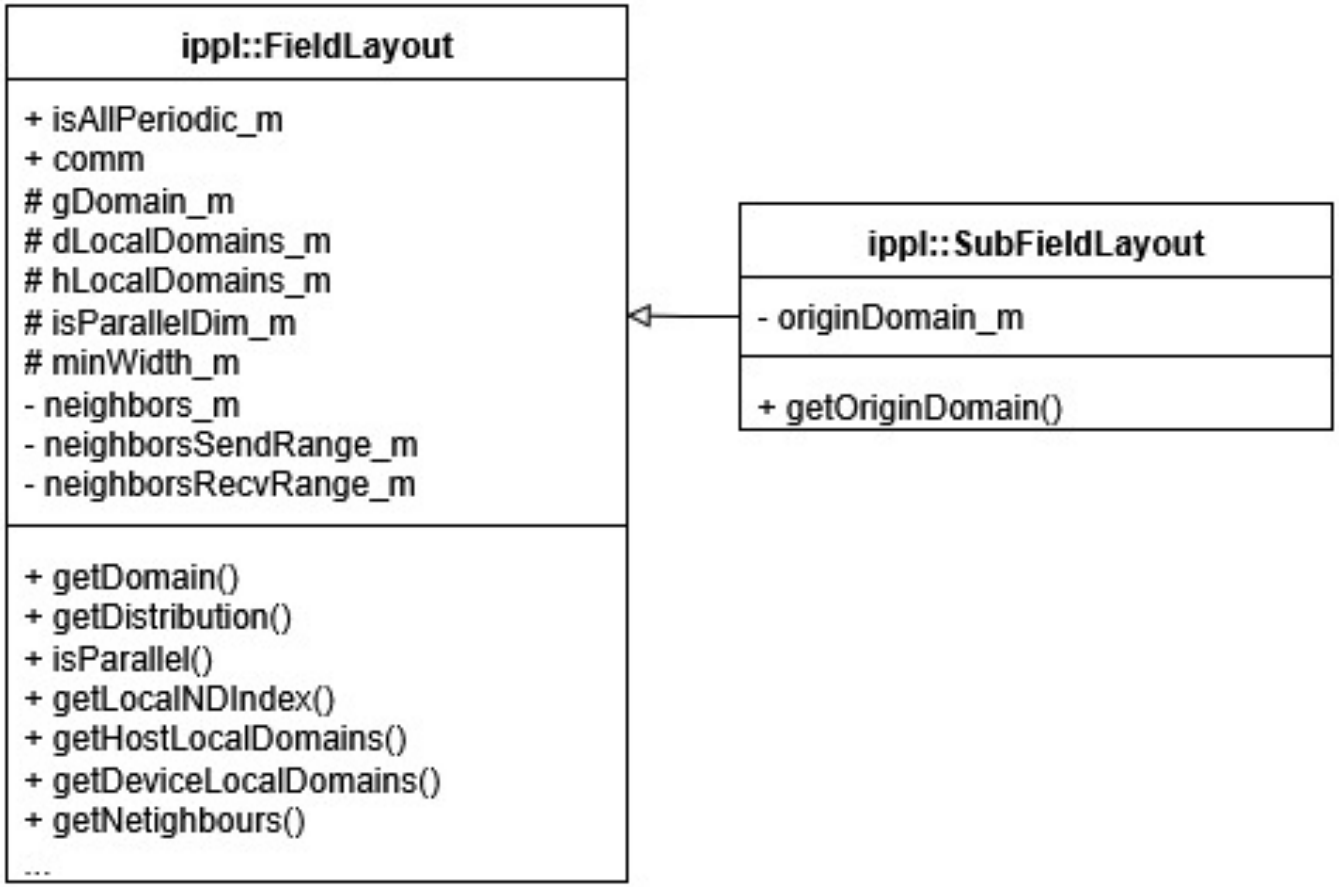
The SubFieldLayout



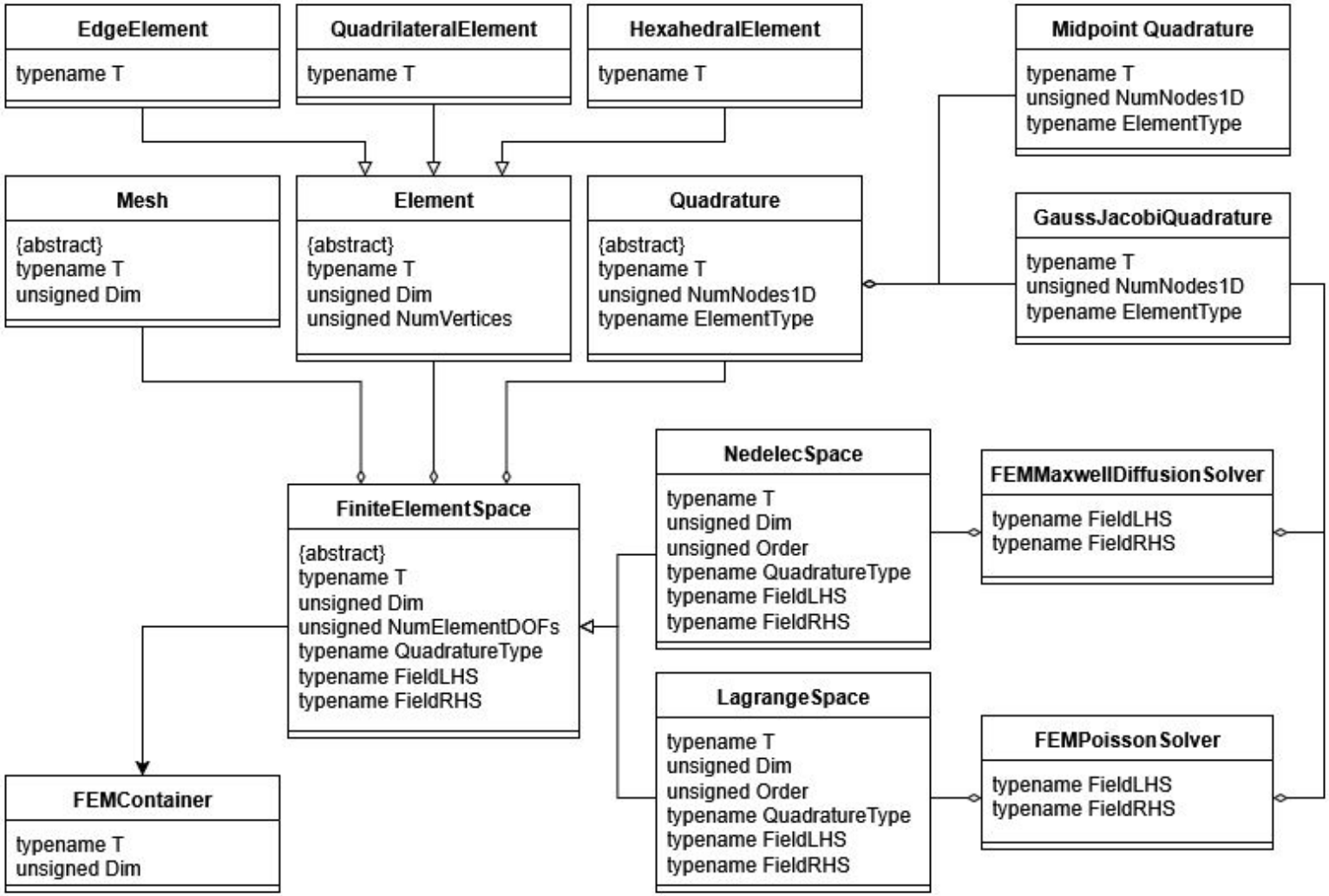
The SubFieldLayout



The SubFieldLayout



The FEM Framework



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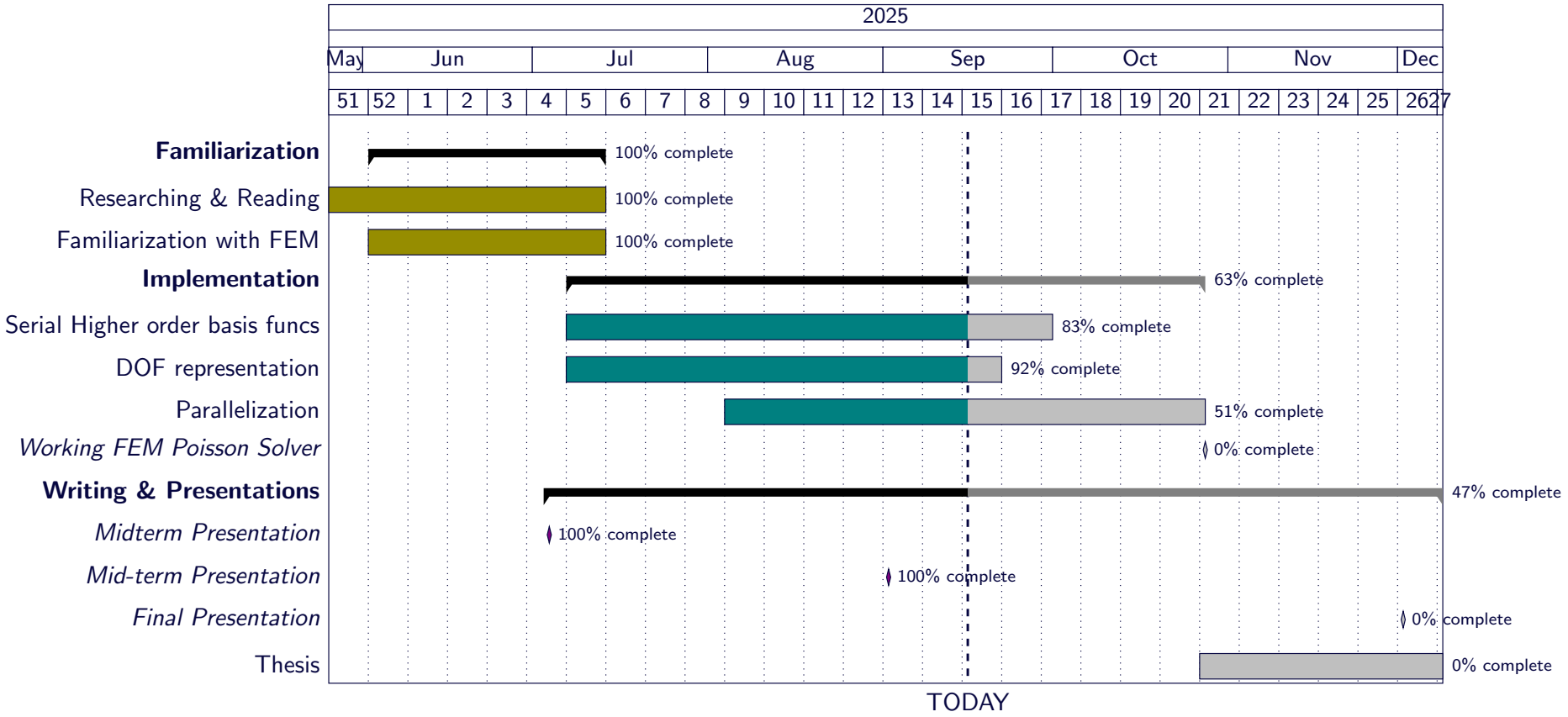
New Timeline



Next Steps

- Currently working on: Migrate Lagrange Space from using Field to using FEMContainer
- Next steps:
  - Recreate results of PoissonSolver using FEMContainer
  - Create pullrequest for FEMContainer
  - Add higher order basis functions to Lagrange Space
  - Implement boundary conditions correctly

References

[Hiptmair NumCSE] R. Hiptmair. Numerical Methods for Computational Science and Engineering. 2019. URL: https://www.sam.math.ethz.ch/ grsam/NCSE20/NumCSE_Lecture_Document.pdf

[Hiptmair NumPDE] R. Hiptmair. Numerical Methods for (Partial) Differential Equations. 2022. URL: https://www.sam.math.ethz.ch/ grsam/NUMPDEFL/NUMPDE.pdf